Question ID b8a225ff

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: b8a225ff

Circle A in the *xy*-plane has the equation $(x+5)^2+(y-5)^2=4$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the *xy*-plane is $(x+5)^2+(y-5)^2=k$, where k is a constant. What is the value of k?

ID: b8a225ff Answer

Correct Answer:
16

Rationale

The correct answer is 16. An equation of a circle in the *xy*-plane can be written as $(x-t)^2+(y-u)^2=r^2$, where the center of the circle is (t,u) , the radius of the circle is r, and where t, u, and r are constants. It's given that the equation of circle A is $(x+5)^2+(y-5)^2=4$, which is equivalent to $(x+5)^2+(y-5)^2=2^2$. Therefore, the center of circle A is $(-5,5)$ and the radius of circle A is 2. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is $(-5,5)$ and the radius of circle B is $2(2)$, or 4. Substituting -5 for t, 5 for u, and 4 for r into the equation $(x-t)^2+(y-u)^2=r^2$ yields $(x+5)^2+(y-5)^2=4^2$, which is equivalent to $(x+5)^2+(y-5)^2=16$. It follows that the equation of circle B in the *xy*-plane is $(x+5)^2+(y-5)^2=16$. Therefore, the value of k is 16.

Question Difficulty:
Hard